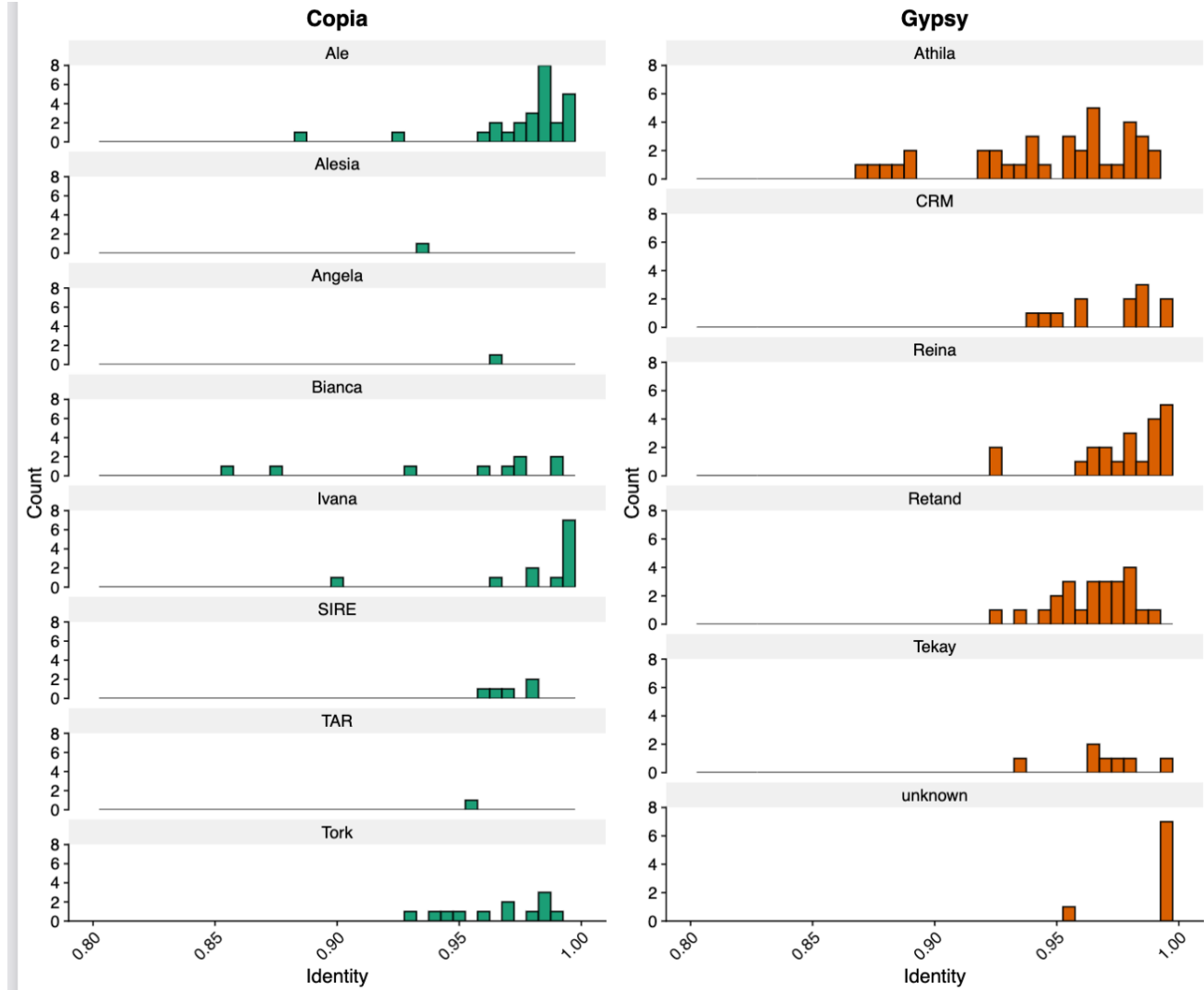


Name: Keke Xia
Accession: Pa-1

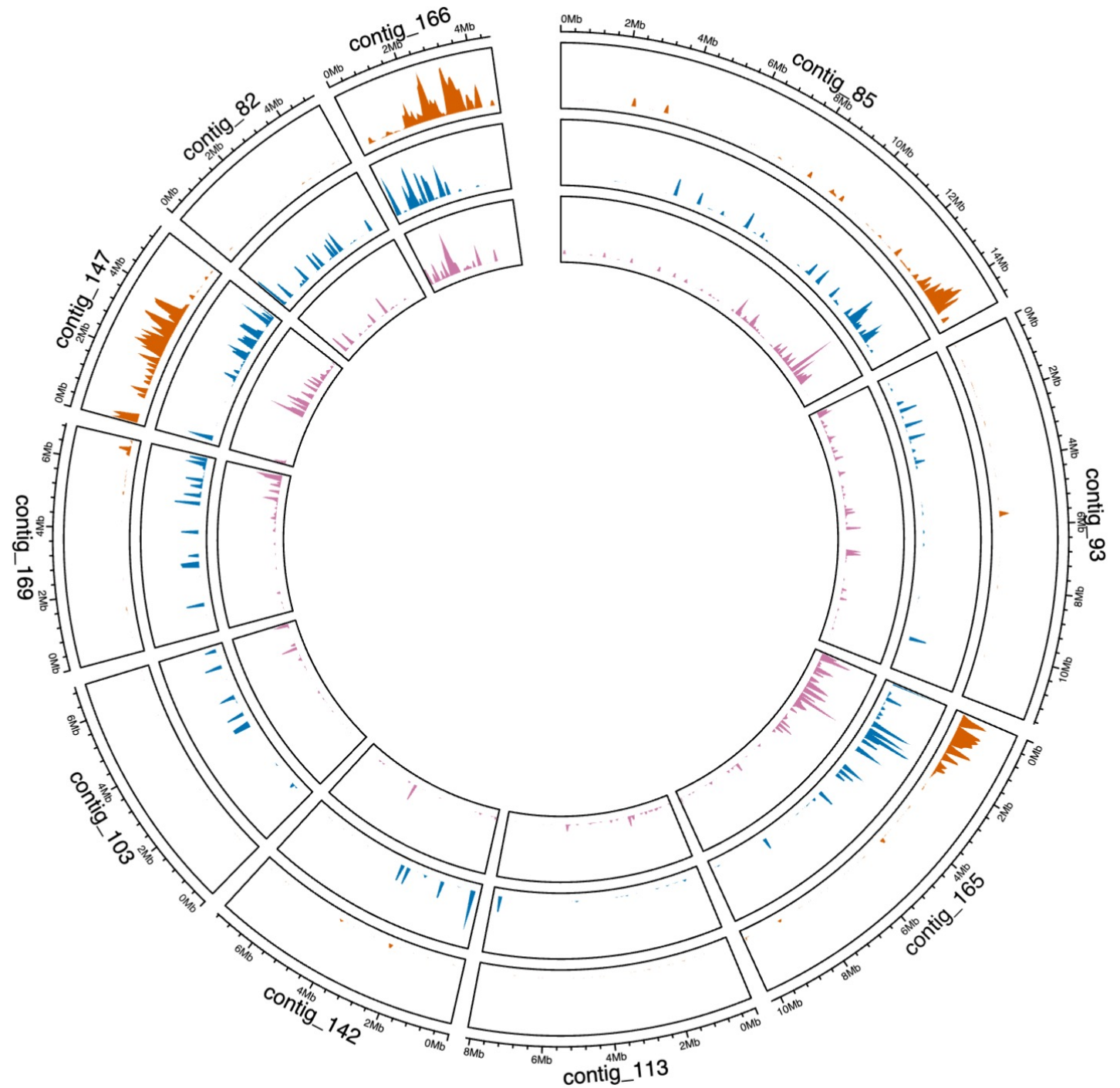
Github Link: https://github.com/keke-xia/organize_annotation_course

Week1-2: Transposable Element Annotation and Classification

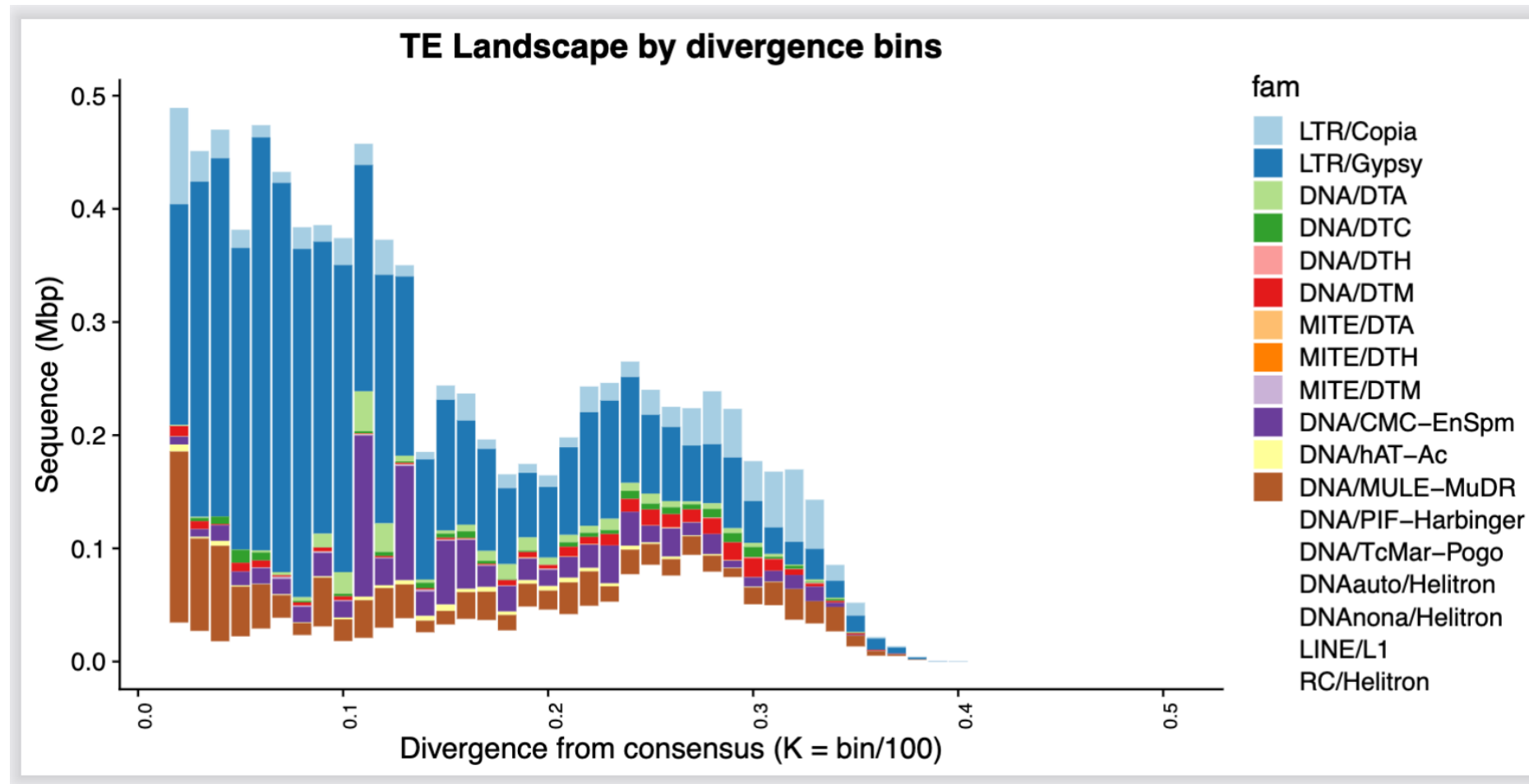
Different clades show very different numbers of full-length LTR-RTs. Copia clades such as Ale and Ivana, and Gypsy clades like Tekay and Reina, contain many young elements with very high identity ($\approx 99\text{--}100\%$), indicating recent activity. In contrast, clades such as Athila and CRM show more diverged copies ($\approx 80\text{--}90\%$), suggesting older insertions.

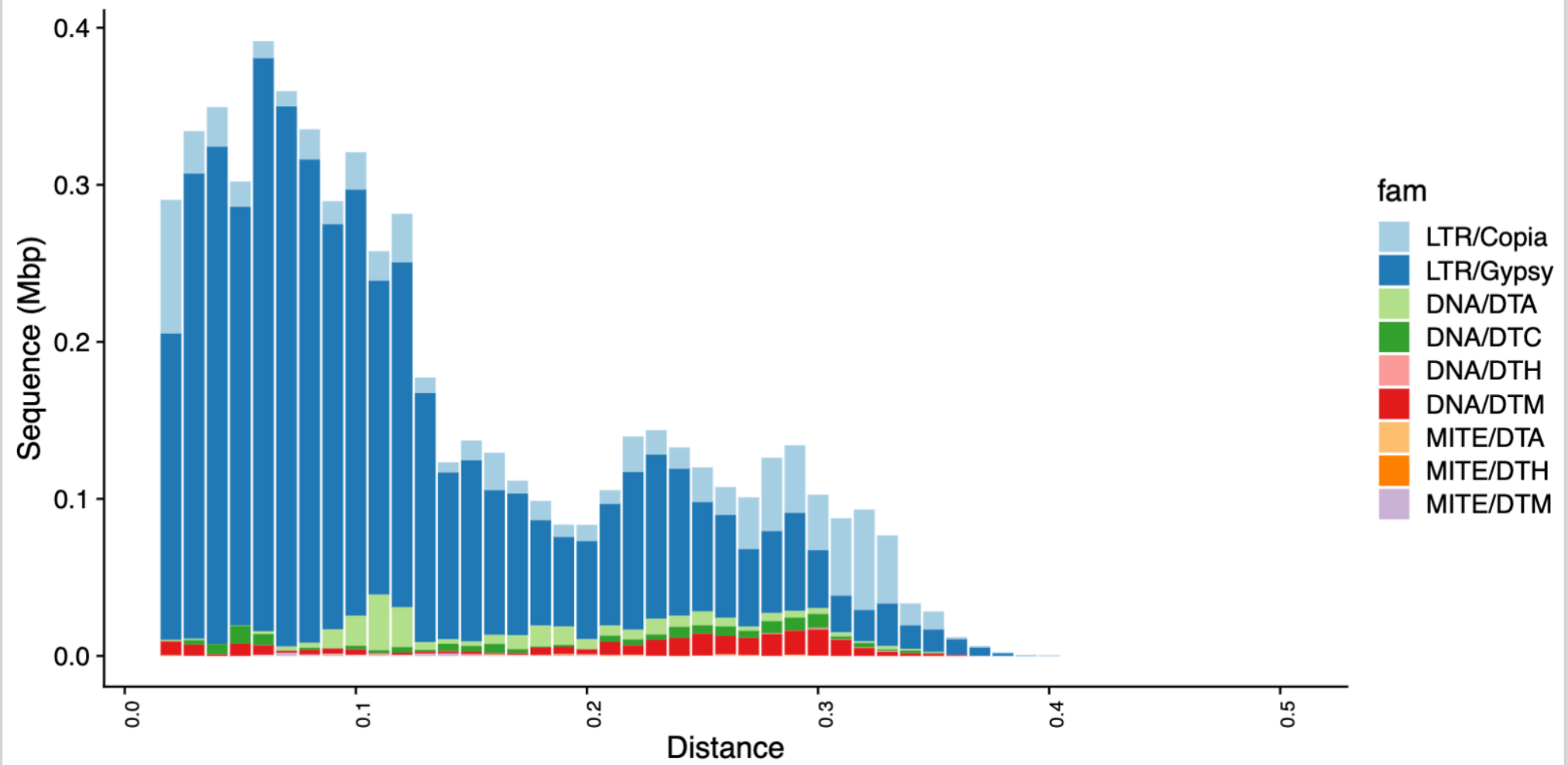


Several contigs contain clear high-density TE clusters, mainly composed of LTR/Gypsy elements. Gypsy shows the strongest and most continuous peaks, whereas Copia is more dispersed and TIR DNA transposons appear only as scattered, low-intensity signals.



Recent and acent activity domained by Copia superfamily

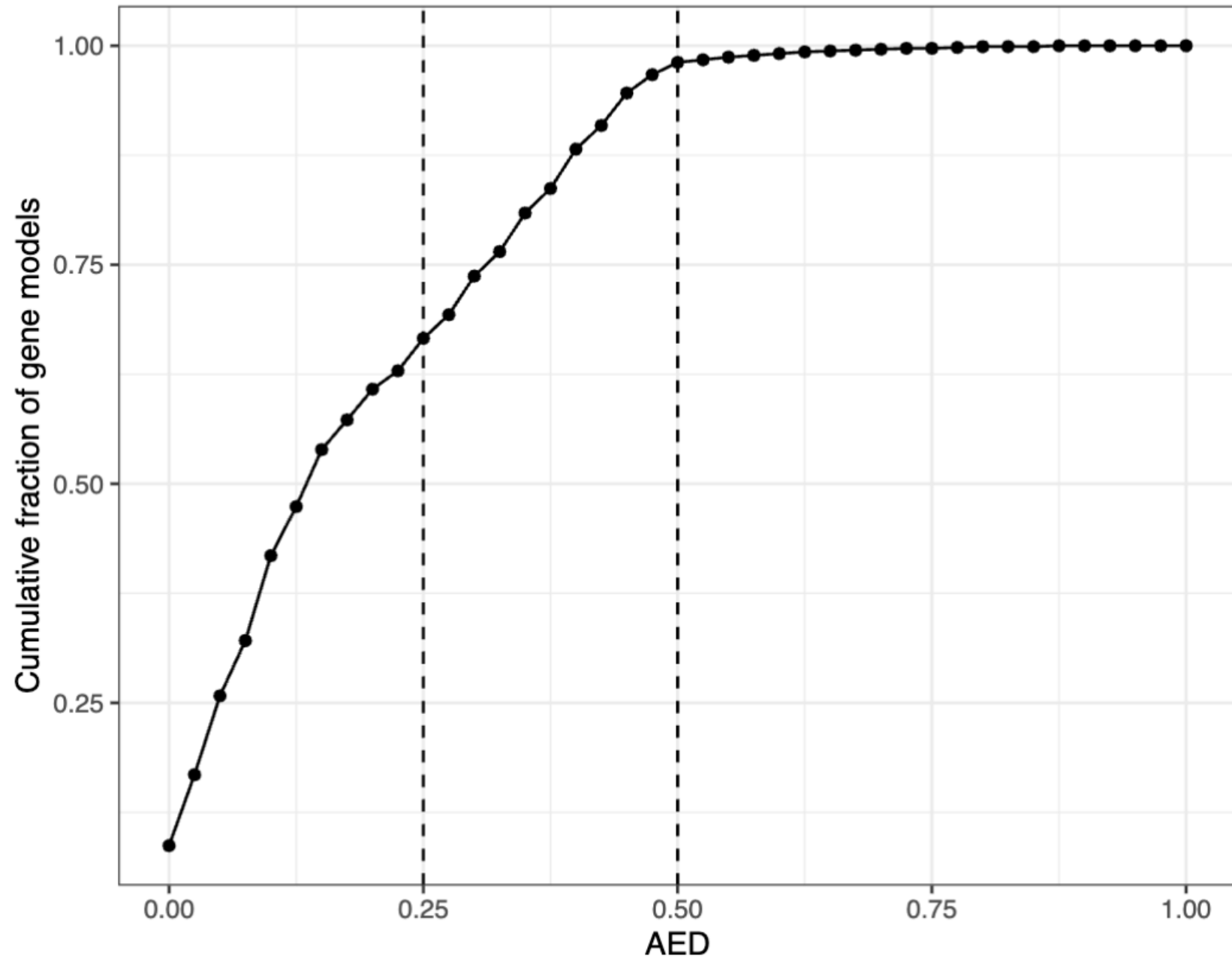




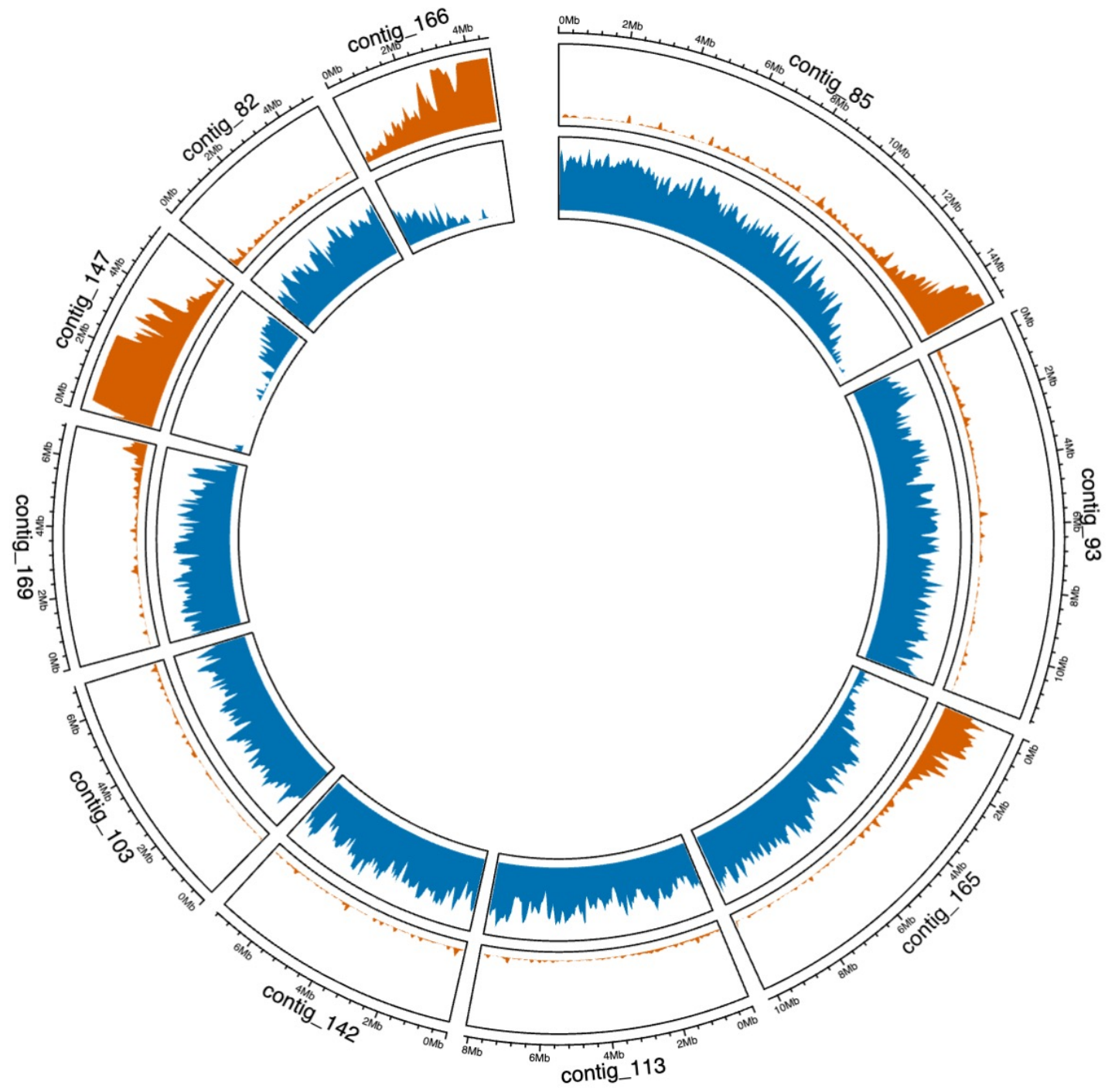
Week3-4: Annotation of genes with the MAKER Pipeline

Cumulative distribution of Annotation Edit Distance (AED)

The AED CDF shows that the majority of gene models are well supported, with ~60% having AED < 0.25 and over 90% having AED < 0.5, indicating high overall annotation quality.

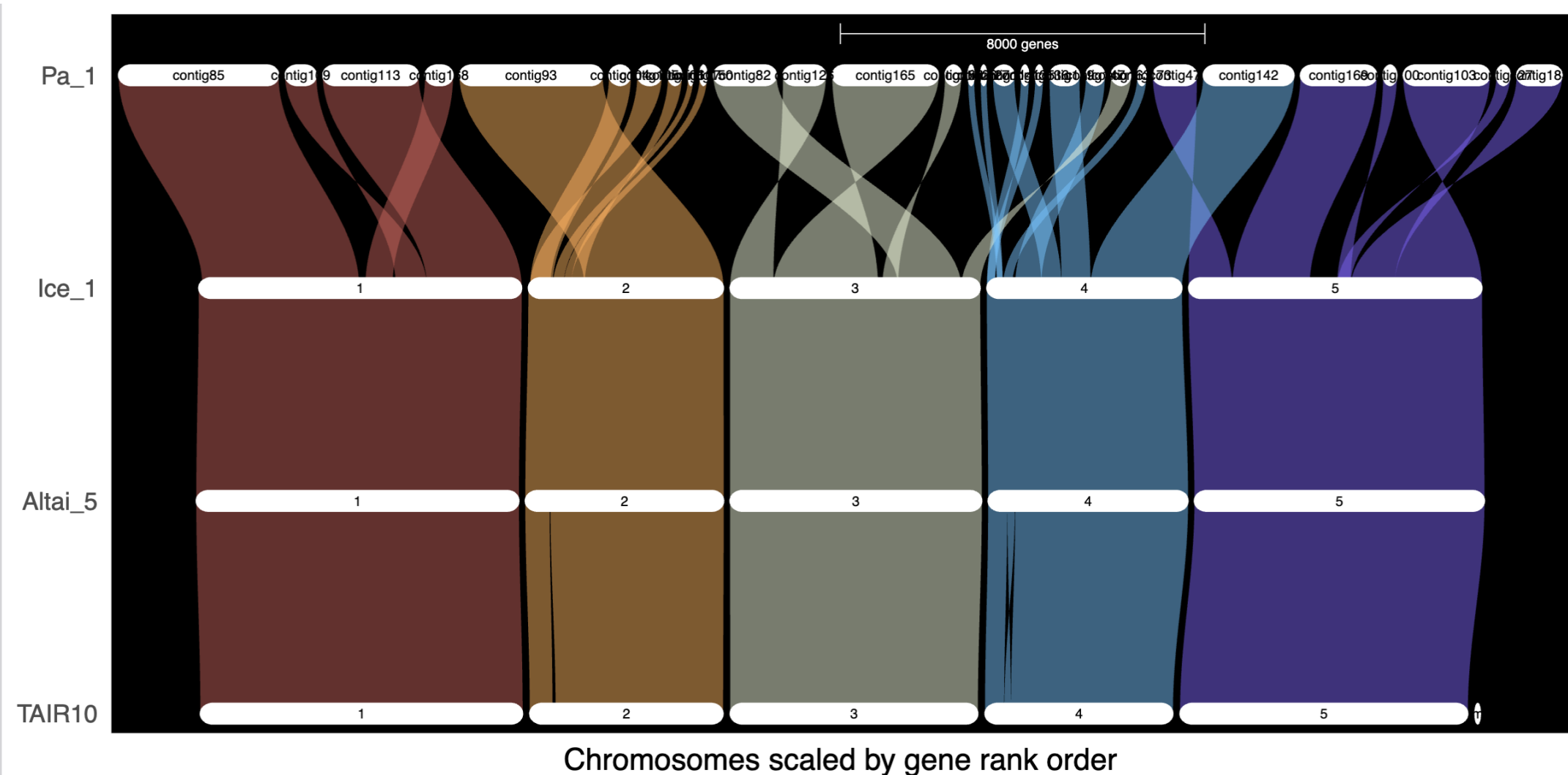


The circos plot shows that TE-rich regions (brown) and gene-rich regions (blue) are generally inversely distributed across the largest scaffolds of the genome, with TE density peaks corresponding to gene-poor regions and vice versa — a canonical pattern in plant genomes.



Week5-6:Orthology based gene functional annotation and genome comparisons

No major chromosomal rearrangements were detected among the accessions; only small local inversions or short-range transpositions are visible, while large-scale synteny remains highly conserved relative to the TAIR10 reference.



Presentation and Data Summary

Accession Info:

Genetic group: European

Specialty :

Three accessions, Lu-1, Pa-1 and Istisu-1, had lower values in the measurement of QV and k-mer completeness. Further alternative allele frequency analysis indicated the presence of heterozygosity, which are unexpected in a pure line, probably due to the mixture of two distinct lineages or segregation of polymorphism in the population.

	Accession	Basepairs covered by TEs	Genes in Accession specific	%_TEs
abnormal	Pa-1	36105639	5029	24.09%
	Others' data	20–24 Mb	700-2000(mostly)	15–17%
normal	Assembler Used	Size of Assembly	Number of Contigs	N50
	Flye	149850558	134	5079500
	Number of TEs	Contig N50 (quickmerge)	Number of Genes	Number of filtered genes
	27578	9672821	29114	29114
	genes with Blast hit	Genes without Blast hits	Genes in core	Contig L50 (quickmerge)
	TAIR10=31,580; Uniprot: 24,063	TAIR10=484; Uniprot: 8,001	25099	6
	Assembly size (quickmerge)	Contig number (quickmerge)	Genes with high quality blastn hit	Basepairs Covered by TEs
	139768900	41	NA	NA